



**COMSATS University Islamabad, Sahiwal  
Campus**

# **SOFTWARE REQUIREMENTS SPECIFICATION**

**(SRS DOCUMENT)**

**for**

**< UniLost – AI-Based University Lost & Found System >**

**Version 1.0**

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## Revision History

| Name | Date | Reason for changes | Version |
|------|------|--------------------|---------|
|      |      |                    |         |
|      |      |                    |         |

## Application Evaluation History

| <b>Comments (by committee)</b><br>*include the ones given at scope time both in doc and presentation | <b>Action Taken</b> |
|------------------------------------------------------------------------------------------------------|---------------------|
|                                                                                                      |                     |
|                                                                                                      |                     |

**Supervised by**  
**<Supervisor's Name>**

Signature \_\_\_\_\_

## Introduction

In the introduction section, an overview of the UniLost system is provided, and the purpose and scope of this Software Requirements Specification document are defined. This section is used as a basis for understanding the functionality and structure of the system.

## **Purpose**

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the UniLost system, its functional and non-functional requirements, its features and constraints, and its interfaces. This document is intended to serve as a reference guide for the developers and testers to develop the system as per the specified requirements. In addition to this, it also serves as a guide to understand the behavior of the system and its performance.

## **Scope**

UniLost is a lost and found management system, especially designed for university students, using the mobile platform. The system offers a centralized platform for users to report lost and found items by providing relevant details such as title, description, category, location, and images.

The system employs Artificial Intelligence, which helps in the automatic analysis and comparison of lost and found items using text and image similarity. Once the system identifies a potential match, it generates a match score and sends real-time notifications to the relevant users.

The system also offers a chat feature, which enables direct communication between the finder and the owner of the lost item, thus verifying the details of the lost item and arranging the return of the item.

The main aim of developing UniLost is to enhance the recovery of lost items, reduce the effort, and offer a reliable, efficient, and user-friendly system for managing lost and found items within the university.

## **Overall description**

### **Product perspective**

UniLost is a standalone mobile application developed using Flutter and Firebase. It replaces informal systems like WhatsApp groups with a structured platform.

### **Operating environment**

OE-1 : The system will operate on Android smartphones

OE-2 : The system will require internet connectivity

OE-3 : Firebase will be used as backend

### **Design and implementation constraints**

CO-1 : System will be developed using Flutter

CO-2 : Firebase must be used

CO-3 : AI accuracy depends on input

CO-4 : Internet required

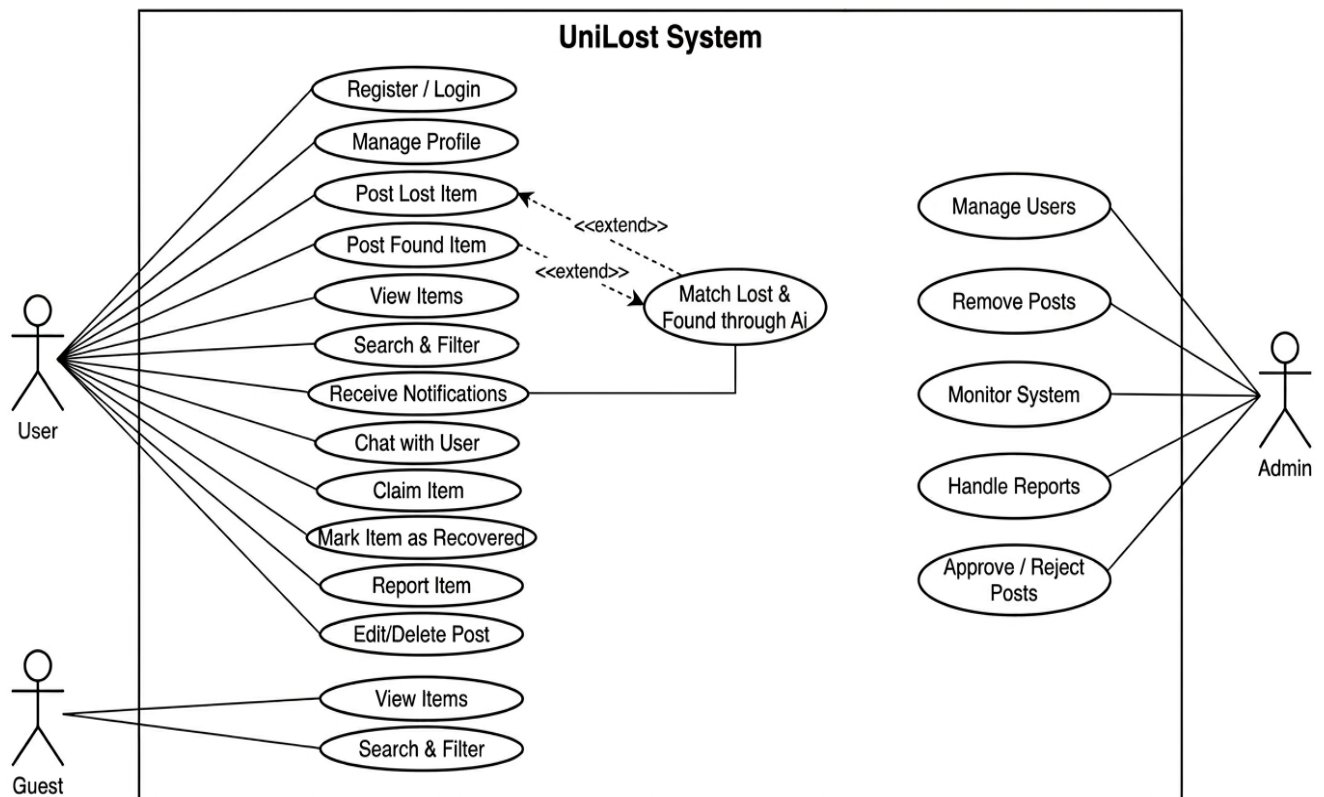
## Requirement identifying technique

The requirements for the UniLost system were identified using the Use Case Technique, which is appropriate for interactive mobile application software. This approach helps to clarify the interaction between the user and the system by creating different scenarios for user interaction with the system.

Use cases were created for the UniLost system to identify the functional requirements of the system, such as user authentication, posting of lost and found items, AI-based matching of the user and the owner of the lost item, sending of notifications, and chat functionality. These use cases are scenarios of interaction between the user and the system.

In addition to the above techniques, the user requirements were also gathered by observing the user and interviewing a few students. The existing systems were also studied for gathering the user requirements for the UniLost system, such as WhatsApp groups and traditional notice boards.

## Use case diagram



## Use case description

Use case descriptions are employed to illustrate the intricate interaction between users and the UniLost system. In essence, each use case describes a specific functionality of the system, including the actors, preconditions, normal flow of events, alternative flows of events, and outcomes.

These use cases are essential in understanding how users will interact with the system and how the system will respond to different conditions. The following tables illustrate the major use cases of the UniLost system.

**Table 5** User Authentication

|                           |                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>       | UC-1                                                                                                                                                                                  |
| <b>Use Case Name:</b>     | User Authentication                                                                                                                                                                   |
| <b>Actors:</b>            | User                                                                                                                                                                                  |
| <b>Description:</b>       | User registers or logs into the system                                                                                                                                                |
| <b>Trigger:</b>           | User opens the application                                                                                                                                                            |
| <b>Preconditions:</b>     | User has internet access                                                                                                                                                              |
| <b>Postconditions:</b>    | User is successfully logged into the system                                                                                                                                           |
| <b>Normal Flow:</b>       | <ol style="list-style-type: none"><li>1. User opens app</li><li>2. Selects login/signup</li><li>3. Enters credentials</li><li>4. System validates</li><li>5. Access granted</li></ol> |
| <b>Alternative Flows:</b> | User selects guest mode → limited access granted                                                                                                                                      |
| <b>Exceptions:</b>        | Invalid credentials → error message displayed                                                                                                                                         |
| <b>Business Rules</b>     | BR-1: Only registered users can access full features                                                                                                                                  |
| <b>Assumptions:</b>       | User provides valid input                                                                                                                                                             |

**Table 5** Post Lost Item

|                       |                |
|-----------------------|----------------|
| <b>Use Case ID:</b>   | UC-2           |
| <b>Use Case Name:</b> | Post Lost Item |

|                     |                        |
|---------------------|------------------------|
| <b>Actors:</b>      | User                   |
| <b>Description:</b> | User reports lost item |

|                           |                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trigger:</b>           | User clicks post                                                                                                                      |
| <b>Preconditions:</b>     | User logged in                                                                                                                        |
| <b>Postconditions:</b>    | Item stored in system                                                                                                                 |
| <b>Normal Flow:</b>       | <ol style="list-style-type: none"> <li>1. Enter details</li> <li>2. Upload image</li> <li>3. Submit</li> <li>4. Data saved</li> </ol> |
| <b>Alternative Flows:</b> | Missing data → validation error                                                                                                       |
| <b>Exceptions:</b>        | Image upload fails                                                                                                                    |
| <b>Business Rules</b>     | BR-2: All fields required                                                                                                             |
| <b>Assumptions:</b>       | Valid data provided                                                                                                                   |

**Table 5** Post Found Item

|                       |                         |
|-----------------------|-------------------------|
| <b>Use Case ID:</b>   | UC-3                    |
| <b>Use Case Name:</b> | Post Found Item         |
| <b>Actors:</b>        | User                    |
| <b>Description:</b>   | User reports Found item |

|                           |                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trigger:</b>           | User clicks post                                                                                                                      |
| <b>Preconditions:</b>     | User logged in                                                                                                                        |
| <b>Postconditions:</b>    | Item stored in system                                                                                                                 |
| <b>Normal Flow:</b>       | <ol style="list-style-type: none"> <li>1. Enter details</li> <li>2. Upload image</li> <li>3. Submit</li> <li>4. Data saved</li> </ol> |
| <b>Alternative Flows:</b> | Missing data → validation error                                                                                                       |

|                       |                           |
|-----------------------|---------------------------|
|                       |                           |
| <b>Exceptions:</b>    | Image upload fails        |
| <b>Business Rules</b> | BR-2: All fields required |
| <b>Assumptions:</b>   | Valid data provided       |

**Table 5** AI Matching

|                       |                            |
|-----------------------|----------------------------|
| <b>Use Case ID:</b>   | UC-4                       |
| <b>Use Case Name:</b> | AI Matching                |
| <b>Actors:</b>        | System                     |
| <b>Description:</b>   | Matches lost & found items |

|                           |                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trigger:</b>           | New item added                                                                                                                          |
| <b>Preconditions:</b>     | Items exist                                                                                                                             |
| <b>Postconditions:</b>    | Match notification sent                                                                                                                 |
| <b>Normal Flow:</b>       | <ol style="list-style-type: none"> <li>1. Extract features</li> <li>2. Compare</li> <li>3. Generate score</li> <li>4. Notify</li> </ol> |
| <b>Alternative Flows:</b> | No match found                                                                                                                          |
| <b>Exceptions:</b>        | AI service failure                                                                                                                      |
| <b>Business Rules</b>     | BR-3: Threshold = 80%                                                                                                                   |
| <b>Assumptions:</b>       | Data available                                                                                                                          |

**Table 5** Notification System

|                       |                                     |
|-----------------------|-------------------------------------|
| <b>Use Case ID:</b>   | UC-5                                |
| <b>Use Case Name:</b> | Notification System                 |
| <b>Actors:</b>        | System                              |
| <b>Description:</b>   | Notify users about matches/messages |

|                           |                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Trigger:</b>           | Match found or message received                                                                         |
| <b>Preconditions:</b>     | User online/offline                                                                                     |
| <b>Postconditions:</b>    | Notification delivered                                                                                  |
| <b>Normal Flow:</b>       | <ol style="list-style-type: none"> <li>1. Event occurs</li> <li>2. System sends notification</li> </ol> |
| <b>Alternative Flows:</b> | Notification delayed                                                                                    |
| <b>Exceptions:</b>        | Network failure                                                                                         |
| <b>Assumptions:</b>       | Internet available                                                                                      |

**Table 5** Chat System

|                       |                               |
|-----------------------|-------------------------------|
| <b>Use Case ID:</b>   | UC-6                          |
| <b>Use Case Name:</b> | Chat System                   |
| <b>Actors:</b>        | Users                         |
| <b>Description:</b>   | Users communicate after match |

|                           |                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Trigger:</b>           | Match notification                                                                                                |
| <b>Preconditions:</b>     | Match exists                                                                                                      |
| <b>Postconditions:</b>    | Messages exchanged                                                                                                |
| <b>Normal Flow:</b>       | <ol style="list-style-type: none"> <li>1. Open chat</li> <li>2. Send message</li> <li>3. Receive reply</li> </ol> |
| <b>Alternative Flows:</b> | No response                                                                                                       |
| <b>Exceptions:</b>        | Message not delivered                                                                                             |
| <b>Assumptions:</b>       | Users cooperate                                                                                                   |

**Table 5** Search Items

|                           |                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>       | UC-7                                                                                                              |
| <b>Use Case Name:</b>     | Search Items                                                                                                      |
| <b>Actors:</b>            | User                                                                                                              |
| <b>Description:</b>       | Search lost/found items                                                                                           |
|                           |                                                                                                                   |
| <b>Trigger:</b>           | User enters keyword                                                                                               |
| <b>Preconditions:</b>     | Items exist                                                                                                       |
| <b>Postconditions:</b>    | Results displayed                                                                                                 |
| <b>Normal Flow:</b>       | <ol style="list-style-type: none"><li>1. Enter keyword</li><li>2. Apply filters</li><li>3. View results</li></ol> |
| <b>Alternative Flows:</b> | No results found                                                                                                  |
| <b>Exceptions:</b>        | Search error                                                                                                      |
| <b>Assumptions:</b>       | Data indexed                                                                                                      |

## Functional Requirements

This section outlines the functional requirements of the UniLost system. The functional requirements are the fundamental characteristics of a system that are needed to ensure it performs the functions it was meant for.

Each of these requirements outlines what the system needs to accomplish or how it needs to behave, including what the user needs to accomplish or how they need to behave. These are grouped according to the different features of the system, such as authentication, posting of items, AI-based matching of items, notifications, and communication.

All the functional requirements are uniquely identified and have a specific format that includes their source, rationale, dependencies, and priority. These are what ensure the system performs correctly both when it is working correctly and when it is faulty.

## Functional Requirement X

**Table 6** User Authentication

|                     |                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>   | FR-1                                                                                                                             |
| <b>Title</b>        | User Authentication                                                                                                              |
| <b>Requirement</b>  | The system shall allow users to register and login using email and password. Invalid credentials shall trigger an error message. |
| <b>Source</b>       | User                                                                                                                             |
| <b>Rationale</b>    | Secure access                                                                                                                    |
| <b>Dependencies</b> | None                                                                                                                             |
| <b>Priority</b>     | High                                                                                                                             |

**Table 6** Guest Access

|                    |                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>  | FR-2                                                                                                                    |
| <b>Title</b>       | Guest Access                                                                                                            |
| <b>Requirement</b> | The system shall allow users to access the application as guests with limited functionality such as viewing items only. |
| <b>Source</b>      | User                                                                                                                    |
| <b>Priority</b>    | Medium                                                                                                                  |

**Table 6** Post Lost Item

|                    |                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>  | FR-3                                                                                                        |
| <b>Title</b>       | Post Lost Item                                                                                              |
| <b>Requirement</b> | The system shall allow users to upload lost item details including title, description, image, and location. |
| <b>Source</b>      | User                                                                                                        |
| <b>Rationale</b>   | Core feature                                                                                                |
| <b>Priority</b>    | High                                                                                                        |

**Table 6** Post Found Item

|                   |                 |
|-------------------|-----------------|
| <b>Identifier</b> | FR-4            |
| <b>Title</b>      | Post Found Item |

|                    |                                                                                      |
|--------------------|--------------------------------------------------------------------------------------|
| <b>Requirement</b> | The system shall allow users to report found items with complete details and images. |
| <b>Source</b>      | User                                                                                 |
| <b>Rationale</b>   | Core feature                                                                         |
| <b>Priority</b>    | High                                                                                 |

**Table 6** AI Matching

|                    |                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------|
| <b>Identifier</b>  | FR-5                                                                                                |
| <b>Title</b>       | AI Matching                                                                                         |
| <b>Requirement</b> | The system shall compare items using text similarity and image analysis and generate a match score. |
| <b>Source</b>      | System                                                                                              |
| <b>Rationale</b>   | Automation                                                                                          |
| <b>Priority</b>    | High                                                                                                |

**Table 6** Notification System

|                    |                                                      |
|--------------------|------------------------------------------------------|
| <b>Identifier</b>  | FR-6                                                 |
| <b>Title</b>       | Notifications                                        |
| <b>Requirement</b> | The system shall notify users when a match is found. |
| <b>Source</b>      | System                                               |
| <b>Rationale</b>   | User awareness                                       |
| <b>Priority</b>    | High                                                 |

**Table 6** Chat System

|                    |                                                                     |
|--------------------|---------------------------------------------------------------------|
| <b>Identifier</b>  | FR-7                                                                |
| <b>Title</b>       | Chat System                                                         |
| <b>Requirement</b> | The system shall allow users to communicate via chat after a match. |
| <b>Source</b>      | User                                                                |
| <b>Rationale</b>   | Communication                                                       |
| <b>Priority</b>    | High                                                                |

**Table 6 Search**

|                    |                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------|
| <b>Identifier</b>  | FR-8                                                                                               |
| <b>Title</b>       | Search                                                                                             |
| <b>Requirement</b> | The system shall allow users to search and filter items based on keywords, category, and location. |
| <b>Source</b>      | User                                                                                               |
| <b>Priority</b>    | Medium                                                                                             |

**Table 6 User Profile**

|                    |                                                                      |
|--------------------|----------------------------------------------------------------------|
| <b>Identifier</b>  | FR-9                                                                 |
| <b>Title</b>       | User Profile                                                         |
| <b>Requirement</b> | The system shall allow users to manage profile and view their posts. |
| <b>Priority</b>    | Medium                                                               |

**Table 6 Chat System**

|                    |                                                                     |
|--------------------|---------------------------------------------------------------------|
| <b>Identifier</b>  | FR-10                                                               |
| <b>Title</b>       | Admin Panel                                                         |
| <b>Requirement</b> | The system shall allow admin to monitor and manage user activities. |
| <b>Source</b>      | User                                                                |
| <b>Priority</b>    | Medium                                                              |

## Non-Functional Requirements

This section identifies the non-functional requirements of the UniLost system, which include the quality attributes of the system, such as usability, performance, security, reliability, and availability, among others.

The non-functional requirements ensure that the system is not only functionally correct but is also efficient, secure, and easy to use. The non-functional requirements are quantifiable and help in the evaluation of the performance of the system.

### Usability

USE-1 : System shall be user-friendly

USE-2 : Easy navigation

## **Performance**

PER-1 : Load within 3 seconds

PER-2 : Real-time notifications

## **Security**

SEC-1 : Secure login

SEC-2 : Data protection

## **Reliability**

REL-1 : No crashes

REL-2 : Data consistency

## **External Interfaces**

### **User Interface**

The UniLost system provides a mobile-based user interface developed using Flutter. The interface is designed to be simple, intuitive, and user-friendly to ensure ease of use for students.

The system includes the following main screens:

- Login and Signup Screen: Allows users to authenticate into the system
- Home Screen: Displays lost and found items
- Post Item Screen: Enables users to report lost or found items
- Chat Screen: Allows communication between users after a match
- Profile Screen: Displays user information and activity

### **Software Interface**

The system interacts with the following software components:

- Firebase Authentication: For user login and signup
- Cloud Firestore: For storing item and user data
- Firebase Cloud Messaging (FCM): For sending notifications
- Firebase Storage: For storing images

### **Hardware Interface**

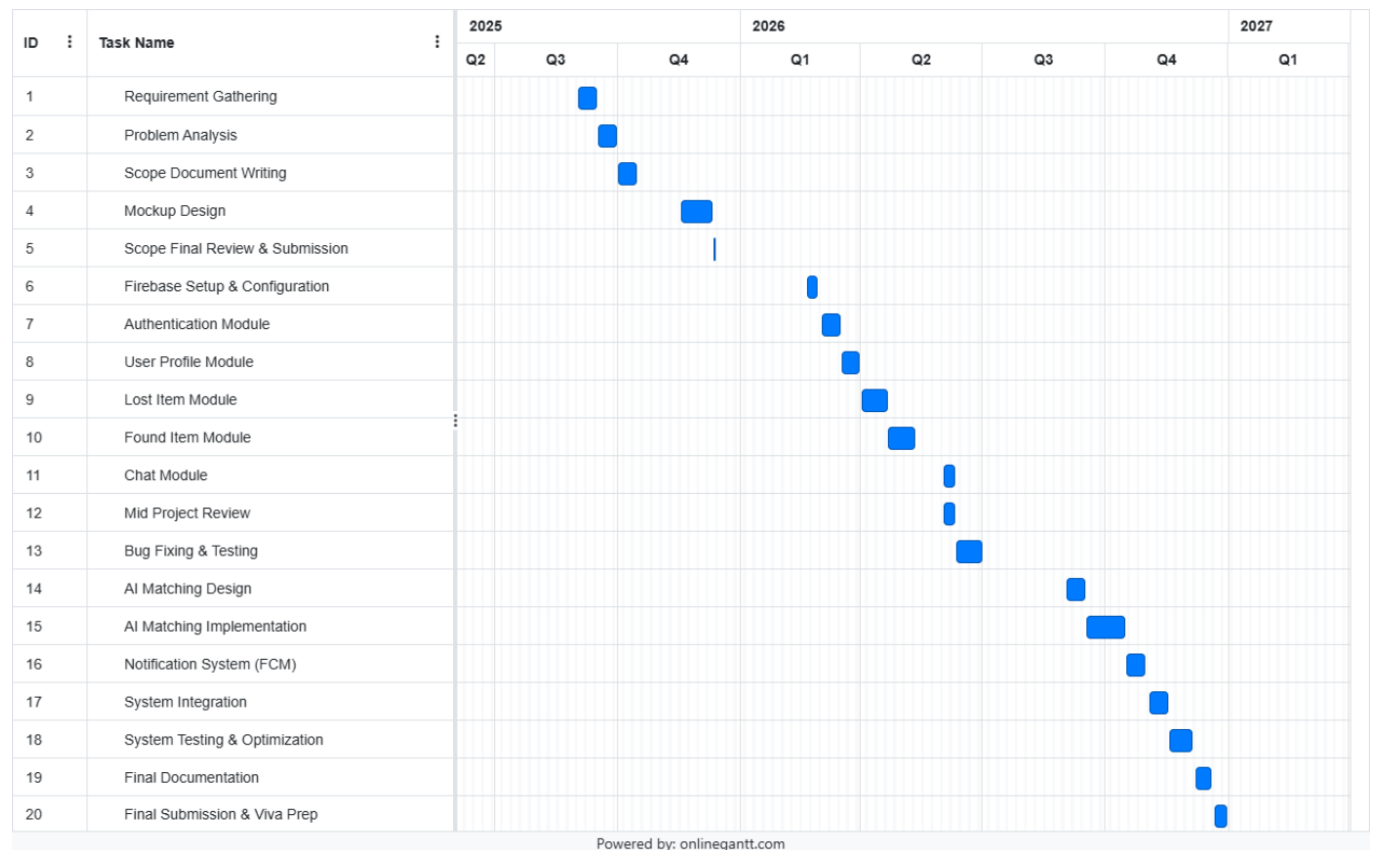
The system requires the following hardware components:

- Android smartphone
- Camera (for capturing and uploading item images)

## Communication Interface

The system requires an internet connection to function properly. Communication between the mobile application and backend services is handled through secure network connections using WiFi or mobile data.

## Gantt Chart



## References

- Flutter Documentation. Flutter – Build apps for any screen. <https://flutter.dev/docs>
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